

Problem Set on Differential Equations

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Problem 1

Find the solution of the initial value problem

$$\dot{a}(t) = ra(t) + s$$

$$a(0) = a_0$$

where r and s are known constants.

Problem 2

Find the solution of the initial value problem

$$\dot{a}(t) = r(t)a(t) + s(t)$$

$$a(0) = a_0$$

where $r(t)$ and $s(t)$ are known functions of time t .

Problem 3

Consider the linear system of differential equations given by

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} \mathbf{x}(t).$$

- A. Find the general solution of the system.
- B. What would you need to find a specific solution of the system?
- C. Draw the trajectories of the system.

Problem 4

Consider the initial value problem

$$\dot{k}(t) = sf(k(t)) - \delta k(t)$$

$$k(0) = k_0$$

where the saving rate $s \in (0, 1)$, the capital depreciation rate $\delta \in (0, 1)$, and the production function f satisfies the *Inada conditions*. That is, f is continuously differentiable and

$$f(0) = 0$$

$$f'(x) > 0$$

$$f''(x) < 0$$

$$\lim_{x \rightarrow 0} f'(x) = +\infty$$

$$\lim_{x \rightarrow +\infty} f'(x) = 0.$$

- A. Give a production function f that satisfies the Inada conditions.
- B. Find the steady state of the system.
- C. Draw the dynamic path of $k(t)$ and show that it converges to the steady state.

Problem 5

The solution of the problem studied in Problem 4 is characterized by a system of two nonlinear first-order differential equations:

$$\begin{aligned}\dot{k}(t) &= f(k(t)) - c(t) - \delta k(t) \\ \frac{\dot{c}(t)}{c(t)} &= \alpha a k(t)^{\alpha-1} - (\delta + \rho).\end{aligned}$$

The first differential equation is the law of motion of capital. The second differential equation is the Euler equation, which describes the optimal path of consumption over time.

- A. Draw the phase diagram of the system.
- B. Linearize the system around its steady state.
- C. Show that the steady state is a saddle point locally.
- D. Suppose the economy is in steady state at time t_0 and there is an unanticipated decrease in the discount factor ρ . Show on your phase diagram the transition dynamics of the model.

Problem 6

Consider the following system of two nonlinear first-order differential equations associated with an investment problem:

$$\begin{aligned}\dot{k}(t) &= \left(\frac{q(t) - 1}{\chi} \right) k(t) \\ \dot{q}(t) &= rq(t) - f'(k(t)) - \frac{1}{2\chi}(q(t) - 1)^2.\end{aligned}$$

The first differential equation is the law of motion of capital $k(t)$. The second differential equation is the law of motion of the costate variable $q(t)$.

- A. Draw the phase diagram.
- B. Show that the steady state is a saddle point locally.

Problem 7

Consider a discrete-time version of the typical growth model:

$$\begin{aligned}k(t+1) &= f(k(t)) - c(t) + (1 - \delta)k(t) \\c(t+1) &= \beta [1 + f'(k(t)) - \delta] c(t).\end{aligned}$$

The discount factor $\beta \in (0, 1)$, the rate of depreciation of capital $\delta \in (0, 1)$, initial capital k_0 is given, and the production function f satisfies the Inada conditions. These two equations are a system of first-order difference equations. Whereas a system of first-order differential equations relates $\dot{\mathbf{x}}(t)$ to $\mathbf{x}(t)$, a system of first-order difference equations relates $\mathbf{x}(t+1)$ to $\mathbf{x}(t)$.

We will see that we can study a system of first-order difference equations with the tools that we used to study systems of first-order differential equations. In particular, we can use phase diagrams to understand the dynamics of the system.

- A. Construct a phase diagram for the system. First, define

$$\begin{aligned}\Delta k &\equiv k(t+1) - k(t), \\ \Delta c &\equiv c(t+1) - c(t).\end{aligned}$$

Second, draw the $\Delta k = 0$ locus and the $\Delta c = 0$ locus on the (k, c) plane. Finally, find the steady state as the intersection of the $\Delta k = 0$ locus and the $\Delta c = 0$ locus.

- B. Show that the steady state is a saddle point in the phase diagram.

Problem 8

We consider the following optimal growth problem. Given initial human capital h_0 and initial physical capital k_0 , choose consumption $c(t)$ and labor $l(t)$ to maximize utility

$$\int_0^{\infty} e^{-\rho t} \ln(c(t)) dt$$

subject to

$$\begin{aligned}\dot{k}(t) &= y(t) - c(t) - \delta k(t) \\ \dot{h}(t) &= b[1 - l(t)]h(t).\end{aligned}$$

Output $y(t)$ is given by

$$y(t) = ak(t)^\alpha [l(t)h(t)]^\beta.$$

We also impose that $0 \leq l(t) \leq 1$. The discount factor $\rho > 0$, the rate of depreciation of physical capital $\delta > 0$, the constants $a > 0$ and $b > 0$, and the production function parameters $\alpha \in (0, 1)$ and $\beta \in (0, 1)$.

- A. Give state and control variables.
- B. Write down the present-value Hamiltonian for this problem.
- C. Derive the optimality conditions.
- D. Show that the growth rate of consumption $c(t)$ is

$$\frac{\dot{c}}{c} = \frac{\alpha y}{k} - (\delta + \rho).$$

- E. From now on, we assume that $b = 0$. Show that $l = 1$ in optimum.
- F. Draw the phase diagram in the (k, c) plane.
- G. Show on the phase diagram that the steady state of the system is a saddle point.
- H. Derive the Jacobian of the system.
- I. Use the Jacobian to show that the steady state of the system is a saddle point.